

Question 1 (Language Models)

Consider the following pre-processed corpus $\mathcal{C}$ with all stopwords removed:

<s> Trump become President US </s>
<s> Obama Trump Biden earlier US President </s>
<s> Trump defeat Harris earlier Hilary </s>

Also consider the pre-processed sentence $\mathcal{S}$:

<s> Biden defeat Trump become US President </s>

- (a) Calculate all the unigram probabilities. Remember that <s> and </s> are also unigrams.
- (b) What is the perplexity of the unigram LM when evaluated over sentence $\mathcal{S}$?
- (c) Calculate all the bigram probabilities for all bigrams in the corpus. Remember that </s><s> is not a valid bigram.

Question 2 (Word embeddings)

You are training a SkipGram word2vec model. Consider a vocabulary of 8 words in this order: **{all, eat, couch, bed, sleep, cats, dogs, peace}**. Each of these words has the same value of in and out weight vectors at a time step, which are as follows (in order):
[2 2 -1], [2 0 1], [-2 2 -2], [1 -1 2], [-1 -1 1], [-1 2 0], [2 -2 -1], [0 1 -2].

Suppose at this time step, the sliding window contains **[all cats sleep couch peace]**, where **sleep** is the centre word and others are context words. We are using a variation of SkipGram, where all context words are considered together as the label / ground truth for calculating the cross-entropy loss. Calculate the **(a)** values at the output before and after softmax, and **(b)** cross-entropy loss at this time step.